

FTD CONSTANTS REFERENCE

Canonical Values to 25-Digit Precision
Computed from $D = 3$ and ϖ

All dimensionless quantities are computed from $D = 3$ and ϖ ; one external unit calibration $K_B = 0.5100 \text{ MeV}$ sets SI units. The Planck mass $m_P = K_B/(\sqrt{2\pi} \cdot \frac{16}{3} \cdot \alpha^{11})$ is derived, not an independent input.

Conventions. $D = 3$ is the spatial dimension (axiom); $\mathcal{D} = N_c N_{\text{base}}^2 - 1 = 47$ is the constraint denominator (derived integer). All masses and couplings use the 4-term corrected α unless tagged “tree.” The tree-level value $\alpha_{\text{tree}} = 1/x_+$ appears only in the Layer 5 table and in r_{eff} , where the uncorrected root is the relevant quantity. $T(n) = n(n+1)/2$ is the n th triangular number.

Layer 0: Transcendental Seed

Symbol	Formula	Value
$\Gamma(1/4)$	Euler gamma function at $1/4$	3.6256099082219083119306851...

Layer 1: Lemniscatic Geometry

Symbol	Formula	Value
ϖ	$2 \int_0^1 \frac{dx}{\sqrt{1-x^4}}^1$	2.6220575542921198104648396...
M	$\frac{\varpi}{\pi} = \frac{1}{\text{AGM}(1, \sqrt{2})}$	0.8346268416740731862814297...

Layer 2: Bridge Constant

Symbol	Formula	Value
G^*	$\frac{\Gamma(1/4)^2}{\sqrt{2} \Gamma(1/2)^2}^2$	2.9586751191886388923108214...

¹Closed form: $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$. The integral definition is π -free.

²Equivalent forms: $G^* = 2\varpi/\sqrt{\pi} = \sqrt{2} \Gamma(1/4)^2/(2\pi)$. The Γ -ratio form is π -free: both $\Gamma(1/4)$ and $\Gamma(1/2)$ are definable by integrals involving only e^{-t} and rational powers of t . The identity $\Gamma(1/2)^2 = \pi$ is a consequence, not an input.

Symbol	Formula	Value
π	$\frac{4\varpi^2}{G^{*2}} = 4\ell$ (derived: not circular ³)	3.1415926535897932384626434...
ℓ	$\frac{\varpi^2}{G^{*2}} = \frac{\pi}{4}$	0.7853981633974483096156608...

Layer 3: Master Quadratic

Equation: $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$

Symbol	Formula	Value
x_+	$8G^{*2}\left(1 + \sqrt{1 - \frac{1}{4G^*}}\right)$	137.03617145815548388160057...
x_-	$8G^{*2}\left(1 - \sqrt{1 - \frac{1}{4G^*}}\right)$	3.0239639163390210039527059...
$x_+ \cdot x_-$	$16 G^{*3}$	414.39243772270942760105875...
$x_+ + x_-$	$16 G^{*2}$	140.06013537449450488555328...
Δ	$64 G^{*3}(4G^* - 1)$	17959.271770230889251899402...

Layer 3b: Precision Formula

$\alpha^{-1} = x_+ - c_1|\varepsilon| + c_2|\varepsilon|^2 - c_3|\varepsilon|^3 - c_4|\varepsilon|^4$

Symbol	Formula	Value
ε	$e^\pi - \pi - 20$	-0.00090002081052423273355702...
c_1	$N_c^2/\mathcal{D} = 9/47$	0.19148936170212765957446809...
c_2	$(N_{\text{eff}} - 2N_{\text{base}})/N_{\text{base}}^3 = 5/64$	0.07812500000000000000000000...
c_3	$N_{\text{base}}/(N_c \cdot \mathcal{D}) = 4/141$	0.02836879432624113475177305...
c_4	$(N_c \cdot \mathcal{D})/(b_3 + N_{\text{base}}) = 141/11$	12.818181818181818181818182...
α^{-1}	(4-term result)	137.03599917700004140583386...

CODATA 2022: $\alpha^{-1} = 137.035\,999\,177(21)$. Agreement: < 0.001 ppt.

Layer 4: Framework Integers

Symbol	Formula	Value
N_c	$\lfloor x_- \rfloor$	3
N_{base}	$2^{(D+1)/2}$	4
b_3	$(11N_c - 2N_f)/3, \quad N_f = 2N_c = 6$	7

³ ϖ is computed by the π -free integral $2\int_0^1 dx/\sqrt{1-x^4}$. G^* is computed by the π -free Γ -ratio above. Their combination gives $\pi = 4\varpi^2/G^{*2}$ without π appearing in any input.

Symbol	Formula	Value
N_{eff}	$b_3 + 2N_c$	13
$\mathcal{D}$	$N_c N_{\text{base}}^2 - 1$	47

Layer 5: Coupling Constants

Symbol	Formula	Value
α_{tree}	$1/x_+$	0.0072973433901380833826792...
α	$1/\alpha_{\text{prec}}^{-1}$ (4-term; used below)	0.0072973525643314228253291...
g_c	$\sqrt{\alpha}$	0.0854245431028543624444677...
$\alpha_s(M_Z)$	$b_3/(b_3 + 4N_{\text{eff}}) = 7/59$	0.1186440677966101694915254...
$\sin^2\theta_W$	$N_c/N_{\text{eff}} = 3/13$	0.2307692307692307692307692...
G_N	$1/(b_3 + N_c)^2 = 1/100$	0.010000000000000000000000...

Layer 6: Mass Scales

Symbol	Formula	Value
m_e/m_P	$\sqrt{2\pi} \cdot \frac{16}{3} \cdot \alpha^{11}$	$4.1774117506818401689118... \times 10^{-23}$
$K_B = m_e$	(lattice natural unit)	1 (= 0.5100 MeV)
K_{genesis}	$N_c \cdot K_B$	3 (= 1.530 MeV)

Layer 7: Mass Ratios (Exact Integers)

Ratio	Formula	Value
m_μ/m_e	$3b_3(b_3 + N_c) - N_c$	207
m_τ/m_e	$(N_{\text{eff}} + N_{\text{base}}) \cdot 207 - 2N_c b_3$	3477
m_p/m_e	$N_{\text{eff}}/\alpha + T(b_3 + N_c), \quad T(10) = 55$	$1836.4679893010005382758...$

Layer 8: Higgs Sector

Symbol	Formula	Value
m_P	$K_B/(\sqrt{2\pi} \cdot \frac{16}{3} \cdot \alpha^{11})$	$2.3938267513054687... \times 10^{22}$ lattice units [†]
v	$m_P \sqrt{2\pi} \alpha^8$	246.07951397853684... GeV
λ_H	$\frac{\sin^2\theta_W}{2 - \sin^2\theta_W} = \frac{3}{23}$	0.1304347826086956521739130...
m_H	$v\sqrt{6/23}$	125.68607601610095... GeV

[†] *Unit conversion.* m_P is computed in lattice units (l.u.), where $K_B = 1$. To obtain GeV values for v and m_H : multiply m_P by $K_B = 0.5100 \times 10^{-3}$ GeV, giving $m_P \approx 1.22085 \times 10^{19}$ GeV; then $v = m_P \sqrt{2\pi} \alpha^8$ and $m_H = v \sqrt{6/23}$ follow in GeV. No additional input enters.

Derived Quantities

Symbol	Formula	Value
W_3	$\frac{\Gamma(1/4)^4}{4\pi^3} = \frac{G^{*2}}{2\pi}$	1.3932039296856768591842463...
c	$1/\sqrt{3}$	0.5773502691896257645091488...
r_{eff}	$\sqrt{x_+} = \sqrt{\alpha_{\text{tree}}^{-1}}$	11.706244976855536653552629...
$1/17$	$1/(N_{\text{eff}} + N_{\text{base}})$	0.0588235294117647058823529...

Confinement (from x_-)

Symbol	Formula	Value
$u_p(x_-)$	$I_1(x_-)/I_0(x_-)$	0.8117414116844551696351218...
$\sigma(x_-)$	$-\ln u_p$	0.2085734480558214996554668...

Key Identities

Identity	Verification	Value
e^π	Gelfond's constant	23.140692632779269005729086...
$e^\pi - \pi - 20$	$= \varepsilon$	-0.00090002081052423273355702...
4ℓ	$= \pi$	3.1415926535897932384626434...
$16G^{*2}$	$= x_+ + x_-$	140.06013537449450488555328...
$16G^{*3}$	$= x_+ \cdot x_-$	414.39243772270942760105875...

All values computed with `mpmath` at 80-digit internal precision; 25 digits displayed.

Canonical source: `scripts/constants.py`.

Engine mirror: `engine/include/ftd/ontic.h` (`constexpr`).